NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-39012/33A
11 January 1973
SUPERSEDING
MIL-C-39012/33
27 February 1967

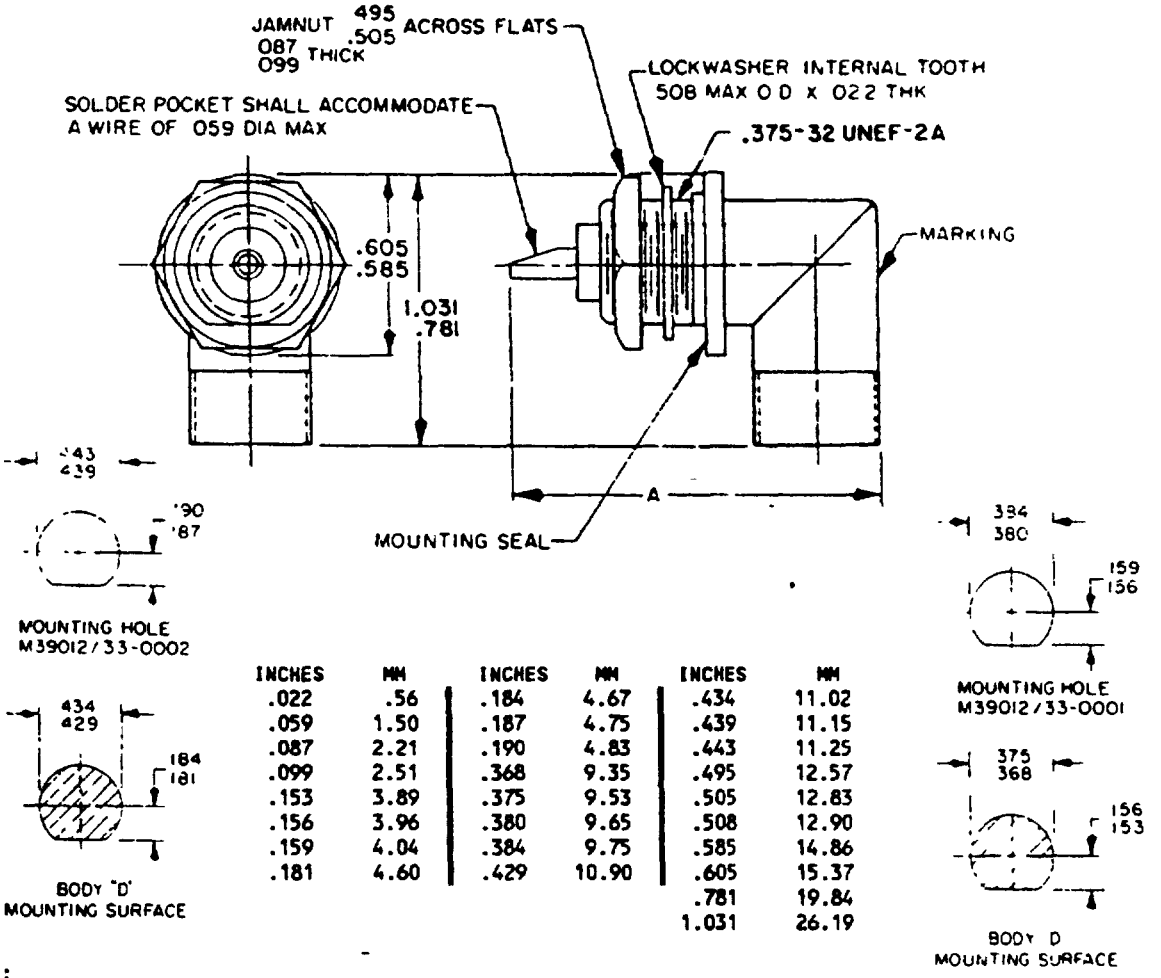
PERFORMANCE SPECIFICATION

CONNECTORS, COAXIAL, RADIO FREQUENCY

(SERIES TNC (UNCABLED - RECEPTACLE, SOCKET, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39012.



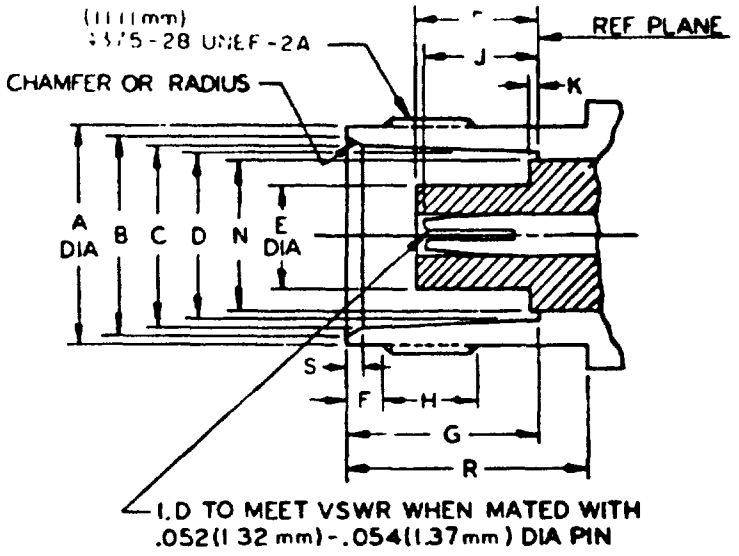
INCHES	MM	INCHES	MM	INCHES	MM
.022	.56	.184	4.67	.434	11.02
.059	1.50	.187	4.75	.439	11.15
.087	2.21	.190	4.83	.443	11.25
.099	2.51	.368	9.35	.495	12.57
.153	3.89	.375	9.53	.505	12.83
.156	3.96	.380	9.65	.508	12.90
.159	4.04	.384	9.75	.585	14.86
.181	4.60	.429	10.90	.605	15.37
				.781	19.84
				1.031	26.19

NOTES:
1. Dimensions are in inches.
2. For dimension A see table I.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. All undimensioned pictoral configurations are for reference purposes only

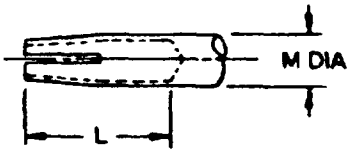
FIGURE 1 General configuration

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:54Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.378 (9.60)	.381 (9.68)
B	.345 (8.76)	.356 (9.04)
C	.327 (8.31)	.333 (8.46)
D	.319 (8.10)	.321 (8.15)
E		.186 (4.72)
F	.068 (1.73)	.088 (2.24)
G	.329 (8.36)	.333 (8.46)
H	.187 (4.75)	
J	1.86 (4.72)	.206 (5.23)
K		.006 (.15)
L	.195 (4.95)	
M	.081 (2.06)	.087 (2.21)
N		.256 (6.50)
P	.188 (4.78)	.208 (5.28)
R	.415 (10.56)	
S	.015 (.38)	.030 (.76)



*N dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension K

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 2 Mating dimensions for socket terminations

Source: https://assist.dla.mil -- Downloaded: 202610-21T16:54Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number and overall dimensions.

Dash No.	Dim	Inches-millimeters 1/ Maximum	Maximum panel thickness
0001	A	1 500 (38. 10)	.250 (6. 35)
0002	A	1 359 (34 52)	. 125 (3. 18)

1/ Millimeters are in parentheses.

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	0001 0002	0001 0002

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:54Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range 0 to 11,000 MHz

Voltage rating 500 volts rms maximum working voltage at sea level 125 volts rms maximum at 70,000 feet.

Temperature rating -65°C to +165°C

REQUIREMENTS

Dimensions and configuration See figure 1

Force to engage and disengage
Longitudinal force - Not applicable
Torque - 2 inch-pounds maximum

Coupling proof torque
Not applicable

Inspection conditions
Coupling torque - 4 to 6 inch pounds

Mating characteristics See figure 2 for dimensions
Center contact (female)
Oversize test pin - .057 diameter minimum (non-closed entry contacts only)
Insertion depth - .125 minimum
Number of insertions - 1

Insertion force test - Steel test pin dia .054 minimum
Test pin finish - 16 microinches
Insertion force - 2 pounds maximum.

Withdrawal force test Steel test pin dia .052 maximum
Withdrawal force - 2 oz minimum
Test pin finish - 16 microinches

Hermetic seal Not applicable

Leakage Connector mounted in mounting hole specified on figure 1 with mating end capped Test applicable to mounting seal only Air pressure - 30 psi
Duration - 30 seconds minimum
Applicable to M39012/33-0001 only

Insulation resistance Method 302, test condition B, MIL-STD-202 5,000 megohms minimum

Center contact retention
6 lbs minimum axial force
5 inch-ounces radial torque minimum

Corrosion (salt spray) Method 101, test condition B, MIL-STD-202

Voltage standing wave ratio (VSWR)
Not applicable

Connector durability 500 cycles at 12 cycles/minute maximum The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	20	25
Outer contact	2	Not applicable

Dielectric withstanding voltage Method 301 of MIL-STD-202 1,500 rms minimum at sea level

Vibration, high frequency Method 204, test condition B, MIL-STD-202

Shock Method 213 of MIL-STD-202, test condition I

Temperature cycling Method 102, test condition C, MIL-STD-202, except test high temperature shall be +200°C

Thermal shock Not applicable

Moisture resistance Method 106 of MIL-STD-202 No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

Corona level
Voltage - 375 volts, minimum
Altitude - 70,000 feet

RF high potential withstanding voltage
Voltage and frequency 1,000 volts rms at 5 MHz.
Leakage current Not applicable

Cable retention force Not applicable

Coupling mechanism retention force
Not applicable

RF leakage Not applicable

Insertion loss Not applicable

Part number M39012/33-
(dash number from table I)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:54Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI, EL
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities
Army - AT, AV, ME
Navy - AS, OS
Air Force - 19

Preparing activity:
Army - EL

Agent
DSA - FS

(Project 5935-1799)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDING OFFICER
U.S. ARMY ELECTRONICS COMMAND
ATTN: AMSEL-TD-S
FORT MONMOUTH, N.J. 07703

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:54Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* - Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:54Z
Check the source to verify that this is the current version before use.